ANIKET KULKARNI

VLSI Design & Technology | RTL Design | Verification | Physical Design

Pune, Maharashtra | Expected graduation: 2027

aniketcoolkarni337@gmail.com | anikett20.github.io | LinkedIn | GitHub

PROFESSIONAL SUMMARY

VLSI Design & Technology undergraduate at APCOER with hands-on experience in Verilog RTL, functional verification, and RTL-to-GDSII implementation. Built a five-stage RV32IM + Zicsr microcontroller SoC with interrupt-driven peripherals and WFI sleep control using OpenLane/SKY130. Practical exposure to Synopsys synthesis and physical design workflows.

TECHNICAL SKILLS

RTL & verification: Verilog, SystemVerilog, Icarus Verilog, VCS, Verdi, testbench development, waveform debugging

Architecture: RISC-V RV32IM + Zicsr, pipelining, hazards/forwarding, CSR/traps, UART, GPIO, timer, SPI

Implementation: OpenLane, SKY130, Yosys, OpenROAD, OpenSTA, KLayout, Magic, Netgen, Design Compiler, IC Compiler II

Design foundations: Logic synthesis, floorplanning, placement, CTS, routing, STA, DRC/LVS, CMOS and digital IC design

SELECTED VLSI PROJECTS

32-Bit RISC-V Microcontroller SoC | RV32IM + Zicsr | OpenLane / SKY130 | Oct 2026

- Designed a five-stage CPU with forwarding and hazard control, a 32-cycle iterative multiply/divide unit, machine-mode CSRs/traps, UART, GPIO, timer, and a four-mode SPI master.
- Implemented synchronous WFI sleep control with a two-cycle writeback drain; verified timer-interrupt wakeup and SPI loopback using Icarus Verilog testbenches.
- Completed a 1 mm² GDSII layout with 21,584 routed standard cells; phase-5 reports show zero Magic/KLayout DRC violations and zero Netgen LVS errors.
- Reduced estimated typical-corner switching power by 28.2% versus phase 4 (9.65 to 6.93 mW) with sleep-control and physical-design changes, under a 20 ns clock constraint.

8-Bit Digital Design | Verilog RTL to GDSII | OpenLane / SKY130 | Apr 2026

- Integrated an eight-operation ALU, an 8 × 8-bit synchronous register file, and a 12-instruction control unit in Verilog; checked RTL behavior with Icarus Verilog and inspected GDSII in KLayout.

4-Bit ALU | Synopsys RTL-to-GDSII Flow | Feb 2026

- Verified opcode-level functionality with VCS/Verdi, synthesized with Design Compiler, and completed floorplanning through routing in IC Compiler II.

RELEVANT EXPERIENCE

Digital Electronics & VLSI Intern | Codec Technologies Pvt. Ltd. | Jul–Aug 2026

- Completed structured training in digital logic, number systems/codes, and core VLSI concepts during a two-month internship.

Technical Assistant Intern | Dynomerk Controls India Pvt. Ltd. | Jan–Feb 2026

- Assisted with industrial control-panel assembly and electronic components; used AutoCAD and automation testing tools for Engine and ABS systems.

EDUCATION

Undergraduate Degree in VLSI Design & Technology | APCOER, Pune | 2023–2027

Anantrao Pawar College of Engineering and Research | Savitribai Phule Pune University

WORKSHOPS, CERTIFICATIONS & TRAINING

Semiconductors for AI & AI for Semiconductors | SemiX AI SIG, IIT Bombay | 17 Oct 2026

Guided reference-design walkthrough: 4 × 4 systolic array (16 PEs), skewed dataflow and golden-model checks; Synopsys dc_shell / ICC2 flow from synthesis through routing.

- 4th Annual SemiX Summit, IIT Bombay, 14–17 Oct 2026: semiconductor entrepreneurship, AI compute, and power electronics.
- Physical Design Using Synopsys — VLSIGuru, Sep 2026; VLSI Specialist Skillset Test — Beep, Aug 2026.
- Semiconductor Manufacturing — CeNSE, IISc, Apr 2026; RTL to GDSII — SemiconTantra, Feb 2026.
- Verilog HDL Fundamentals — Udemy, Oct 2025; AI PC Workshop — SemiX, IIT Bombay, Sep 2025.

LEADERSHIP

Technical Symposium Coordinator, SEMICONIX – APCOER: co-organized departmental quizzes and presentations.